Trust & Safety: Assignment 3

Donations on Bluesky

Motivation

Donations are essential for mobilizing support during crises, yet their effectiveness relies on trust. On Bluesky, donation and charity scams have increasingly exploited this trust, especially during humanitarian emergencies when users are most vulnerable and verification is most difficult. Malicious actors use untraceable payment links, QR codes, and impersonation tactics to solicit money that never reaches intended recipients. These scams cause both financial and emotional harm, while simultaneously diverting funds away from legitimate causes. Bluesky's decentralized architecture and rapid content circulation amplify these risks, as new accounts can quickly spread misleading posts before moderation tools can respond. Ensuring the authenticity of donation requests is thus necessary not only for user protection but also for preserving long-term trust to the platform.

Summary of Approach & Policy

While LLM and API based moderation could improve coverage, it raises cost and efficiency challenges. Thus, we propose a lightweight, rule and text-matching based donation-scam detection mechanism that automatically identifies donation-related posts, extracts and classifies any payment endpoints, evaluates the trustworthiness of associated accounts and domains, and produces clear, interpretable labels for users. The policy distinguishes between trustworthy, potentially risky, and unverifiable donation posts through multi-stages: (1) intent detection, (2) payment mechanism extraction, (3) domain and account verification, and (4) final scam-risk assessment. Rather than removing content outright, the system provides structured labels: *no*, *unsure*, *unknown*, to surface risk without over-policing legitimate fundraising. This approach aims to reduce donor harm, elevate credible donation pathways, and strengthen the overall integrity of the donation ecosystem on Bluesky.

Test Description

To evaluate the donation-labeling pipeline, we constructed a test dataset of 163 Bluesky posts using a combination of keyword-based sampling (e.g., "donate") and automated pulls, including random pulls for negatives and targeted pull from recognized charity accounts. We labeled more positives and hard cases over negatives to stress test our labeler. Each post was manually annotated across 6 fields, with "*donation_related*" (whether the post includes donation info), "*contains_payment_mechanism*" (whether the post includes a payment method and which type), and "*scam*" (a preliminary risk assessment based on source verifiability, payment method trustworthiness, and contextual cues), serving as ground-truth targets. Additional labels support interpretation of ambiguous cases.

<table>
<tr><th>Column</th><th colspan="2">donation_related</th><th colspan="2">contains_payment_mechanism</th><th colspan="2">scam</th><th>domain</th><th>verified_org</th><th>urgent_ language</th></tr>
<tr><td rowspan="8">Schema & Count</td><td rowspan="4">yes</td><td rowspan="4">94</td><td rowspan="2">payment link</td><td rowspan="2">3</td><td rowspan="4">no</td><td rowspan="4">109</td><td>cryptocurrency</td><td rowspan="3">yes</td><td rowspan="4">yes</td></tr>
<tr><td>war</td></tr>
<tr><td rowspan="2">payment handle</td><td rowspan="2">5</td><td>poverty</td></tr>
<tr><td>education</td><td rowspan="3">no</td></tr>
<tr><td rowspan="4">no</td><td rowspan="4">69</td><td rowspan="2">fundraising website</td><td rowspan="2">60</td><td rowspan="4">unsure</td><td rowspan="4">43</td><td>politics</td><td rowspan="4">no</td></tr>
<tr><td>disaster relief</td></tr>
<tr><td>qrcode</td><td>3</td><td>animal</td><td rowspan="2">unsure</td></tr>
<tr><td>no</td><td>81</td><td>other</td></tr>
</table>

The pipeline was executed end-to-end. Accuracy, precision, recall, and F1 were computed, and the system was stress-tested on edge cases involving spelling distortions, multi-modal posts (embedded URLs, facet links, QR codes), and non-standard payment endpoints. This test design is appropriate because it evaluates multi-modal input pathways and edge cases, and exposes the system to varying levels of verification and domain trust. The size (>150) also aligns with the requirement to emit labels at scale.

Test Results & Discussions

Overall performance was strong on the foundational tasks. Donation intent, payment detection, and mechanism classification all achieved over 90% accuracy. Scam-risk evaluation (computed only on “no” vs. “unsure") reached 84.83% accuracy.

Task	Correct	Wrong	Accuracy
donation_related?	139	13	91.54%
has_payment?	139	13	91.54%
payment_mechanism?	139	13	91.54%
scam? (“unknown” ignored)	123	22	84.83%

The donation-intent classifier shows a strong recall, meaning it rarely missed genuine donation posts and most errors were mild false positives caused by generic fundraising words. Payment-mechanism detection was highly precise but less complete, with a recall of 0.82. Scam-risk classification was the weakest component, particularly for false negatives. Across 145 labeled cases, the model achieved precision 0.84 and recall 0.52, showing the difficulty of surfacing potential scams when verification signals are sparse and rule-based logic is not sufficiently strict. One clear example is *ActBlue*: our charity-sites database categorizes ActBlue as “verified” because it is the official donation processor for the U.S. Democratic campaigns. As a result, all ActBlue endpoints were automatically labeled “not risky.” However, ActBlue has documented fraud cases, and several ActBlue posts were manually labeled as “unsure” in our dataset. This mismatch shows limitations: relying solely on organization and account level verification can mask legitimate risk, leading the model to under-detect suspicious cases tied to well-known but imperfect platforms.

Donation Related	Pred Related	Pred Not Related	**Recall 1**	**Has Payment**	Pred Yes	Pred No	**Recall 2**	**Scam**	Pred Not Scam	Pred Risky	**Recall 3**
Actual Related	83	2	0.98	Actual Yes	58	13	0.82	Actual Not Scam	101	2	0.98
Actual Not Related	11	56	0.84	Actual No	0	81	1	Actual Risky	20	22	0.52
Precision 1	0.88	0.97	F1 = 0.93	**Precision 2**	1	0.86	F1 = 0.89	**Precision 3**	0.84	0.92	F1 = 0.67

Stress Test & Performance

To evaluate robustness, we conducted a stress test combining batch execution and targeted edge-case analysis. First, we ran the pipeline on the test set (152 valid entries out of 163) and observed stable performance with a total runtime of 133 seconds (≈0.88s per sample). This demonstrates that our architecture is lightweight and computationally efficient enough for near-real-time moderations. We then examined failure modes through curated edge cases. For Module 1, we included posts with creative spellings (e.g., d0nate), and contextual references (e.g., “donate for toys”, irrelevant to charity), confirming the need for fuzzy matching and context checks. For Module 2, we tested varied payment formats: embedded links, faceted links, email handles, and QR codes, and identified that unclassified QR-decoded URLs account for most false negatives, probably because those QR codes are embedded in images. We also evaluated Module 3 with ambiguous and unseen donation domains, highlighting gaps in our domain database and the need for external threat intelligence (e.g., WebRisk API). Overall, the stress test validates that the system is operationally reliable at scale, while the edge cases pinpoint clear areas for future improvement, particularly in context understanding, QR-code URL classification, and expanding domain-level verification coverage.

Next Steps & Use Cases, and Ethical Implications

Our next steps focus on improving recall without increasing false positives, which is essential for a use case that must protect donors while preserving access for legitimate fundraisers. We plan to incorporate richer contextual and behavioral signals, such as posting history, engagement patterns, and trusted-user endorsements to better distinguish genuine campaigns from opportunistic or identity-shifting scams. Ethically, addressing global equity is important: the labeler currently operates only in English and relies heavily on U.S.-centric verification sources, creating disadvantages for grassroots and non-Western fundraising practices. To support a broader set of users, we intend to expand multilingual coverage and diversify verification pathways beyond large international nonprofits. These steps will make the system more robust, more globally representative, and better suited for real-world donation ecosystems.